Salesforce Project Documentation

RePlastix Innovations: Transforming Plastic Waste into Sustainable Solutions

Implementation Report Based on Completed Work

Project Information	Details
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Environment	Salesforce
Project	RePlastix Innovations
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Scope of this report: the Salesforce components configured, automated, tested, and documented as part of the RePlastix Innovations project.

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PHASE 1 — REQUIREMENT ANALYSIS & PLANNING

1.1 Business Objectives

RePlastix Innovations is designed to provide a centralized Salesforce solution for managing the operational lifecycle of plastic waste and recycled products. The primary objective is to organize collection records, recycling-center information, product inventory, customer orders, and restock requests in one connected system.

The project also aims to reduce manual follow-up by introducing automated low-stock monitoring and programmatic inventory processing. The system is intended to improve visibility of stock levels, prevent invalid transactions, and support timely replenishment when customer demand exceeds available product stock.

1.2 Functional Scope

The implemented scope covers the major records and processes required for the project. The solution uses five custom objects and their relationships to represent the business process. In the org, each object carries the "Re Plastic Innovations" label prefix: Order, Plastic (Plastic Waste), Recycled Product, Recycling Center, and Restock Request.

- Re Plastic Innovations Plastic (Plastic Waste) — records collected plastic waste, including Weight, Type, Collection Date, Status, Location, and the Recycling Center lookup.
- Re Plastic Innovations Recycling Center — stores recycling-center details: Location (geolocation) and processing Capacity.
- Re Plastic Innovations Recycled Product — maintains product Stock Level, Threshold, Price, and the calculated Stock Low On Product formula field.
- Re Plastic Innovations Order — records customer orders with the Customer (Account lookup), selected Recycled Product, Quantity, and Delivery Date.
- Re Plastic Innovations Restock Request — records shortages with the Recycled Product lookup, Requested Quantity, and restock Status.
- Automation covers daily low-stock task creation, order-based stock processing, restock-request creation, approved restock processing, and email notification.

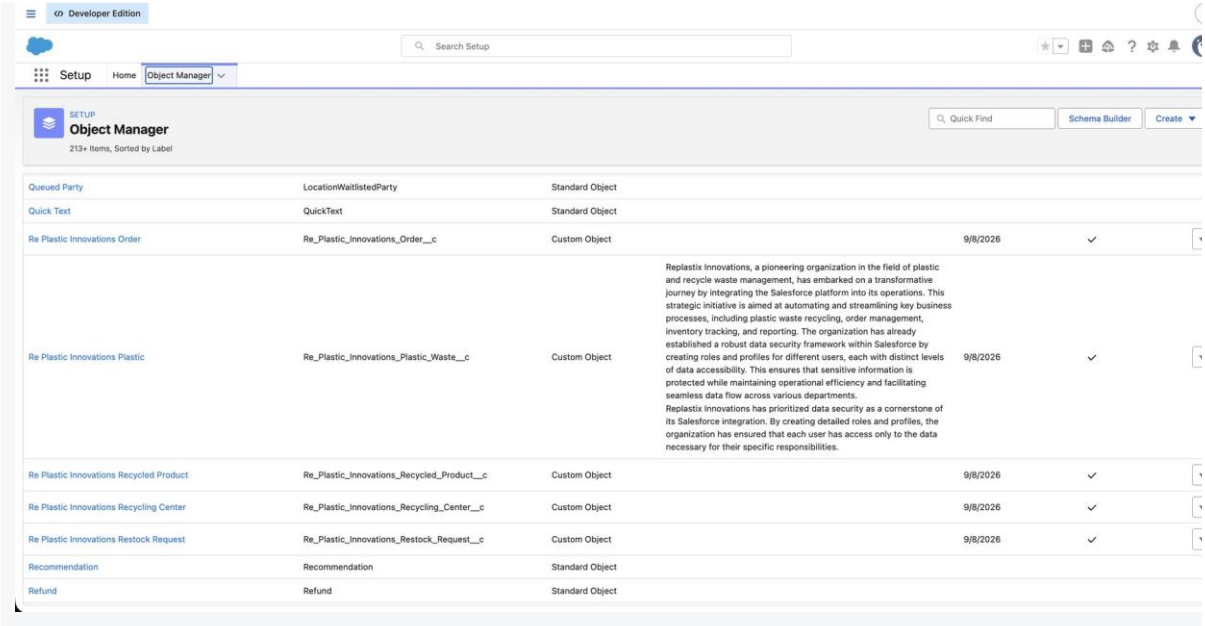


Figure 1: Object Manager showing the five Re Plastic Innovations custom objects

1.3 Stakeholder Mapping

The implementation separates responsibilities through Salesforce roles and profiles. The hierarchy and object permissions are designed so that operational users receive the access required for their work while sensitive records remain restricted by default.

Stakeholder / Role	Primary Responsibility	Relevant Access
CEO	Overall oversight	Higher-level record visibility
Recycling Manager	Plastic-waste operations	Plastic Waste create/read; shared waste records
Sales Representative	Customer orders and products	Order and Account create/read; product read
Warehouse Supervisor	Inventory and restocking	Custom-object create/read/edit; restock visibility

1.4 Execution Roadmap

The implementation was completed in a sequence that moved from data design to application configuration, automation, programmatic processing, security, and testing. This order reduced dependency problems because relationships and fields were established before automation and Apex logic were tested.

- Define business requirements and identify core Salesforce objects.
- Create custom objects, fields, picklists, relationships, and tabs.
- Configure the Lightning application and navigation.
- Implement formula and validation rules for business data quality.
- Create scheduled Flow automation for low-stock monitoring.
- Implement Apex inventory processing and triggers for orders and restock approvals.
- Configure profiles, roles, private record access, and sharing rules.
- Create and execute the required test scenarios and verify successful results.

PHASE 2 — SALESFORCE DEVELOPMENT – BACKEND & CONFIGURATIONS

2.1 Data Architecture

The data model uses separate custom objects for the main business entities. Lookup relationships connect records without combining unrelated responsibilities into one object. This structure supports easier maintenance and allows users to navigate from a product to related orders or restock requests.

Object	Important Fields	Relationship
Re Plastic Innovations Plastic (Plastic Waste)	Weight, Type, Collection Date, Status, Location	Lookup to Recycling Center
Re Plastic Innovations Recycling Center	Location, Capacity	Referenced by Plastic
Re Plastic Innovations Recycled Product	Stock Level, Threshold, Price, Stock Low On Product (formula)	Referenced by Order and Restock Request
Re Plastic Innovations Order	Customer, Recycled Product, Quantity, Delivery Date	Lookup to Account and Recycled Product
Re Plastic Innovations Restock Request	Recycled Product, Requested Quantity, Status	Lookup to Recycled Product

The screenshots below show the Details page and the Fields & Relationships page for each of the five custom objects, including the API names, field data types, and lookup relationships.

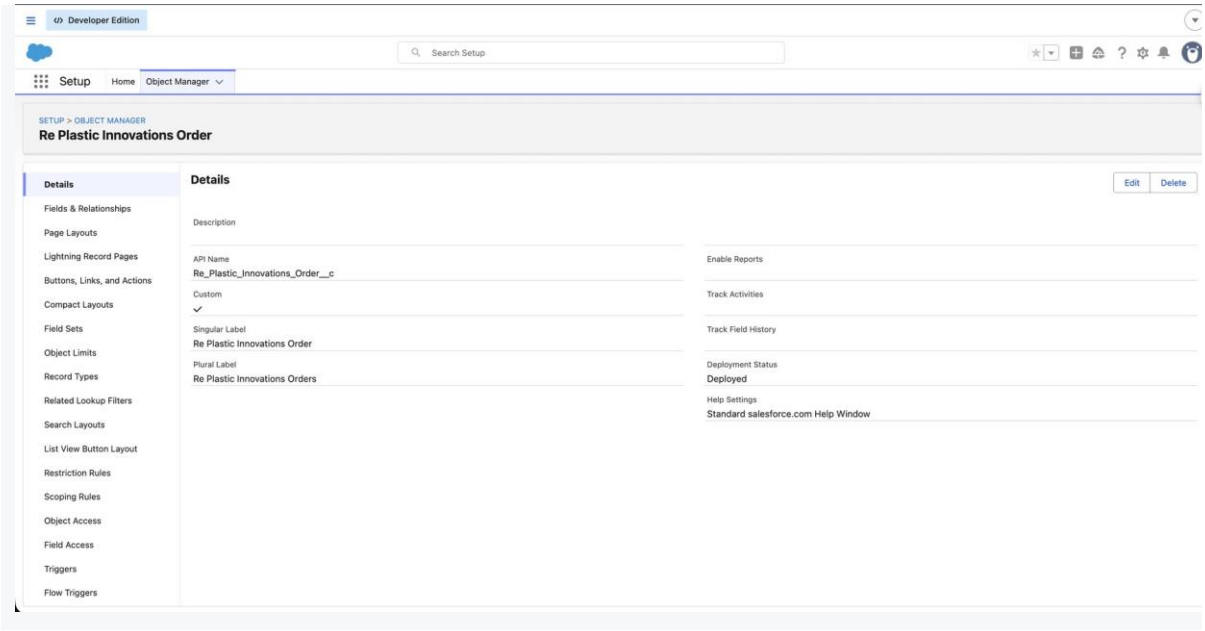


Figure 2: Re Plastic Innovations Order: object details (API name Re_Plastic_Innovations_Order__c)

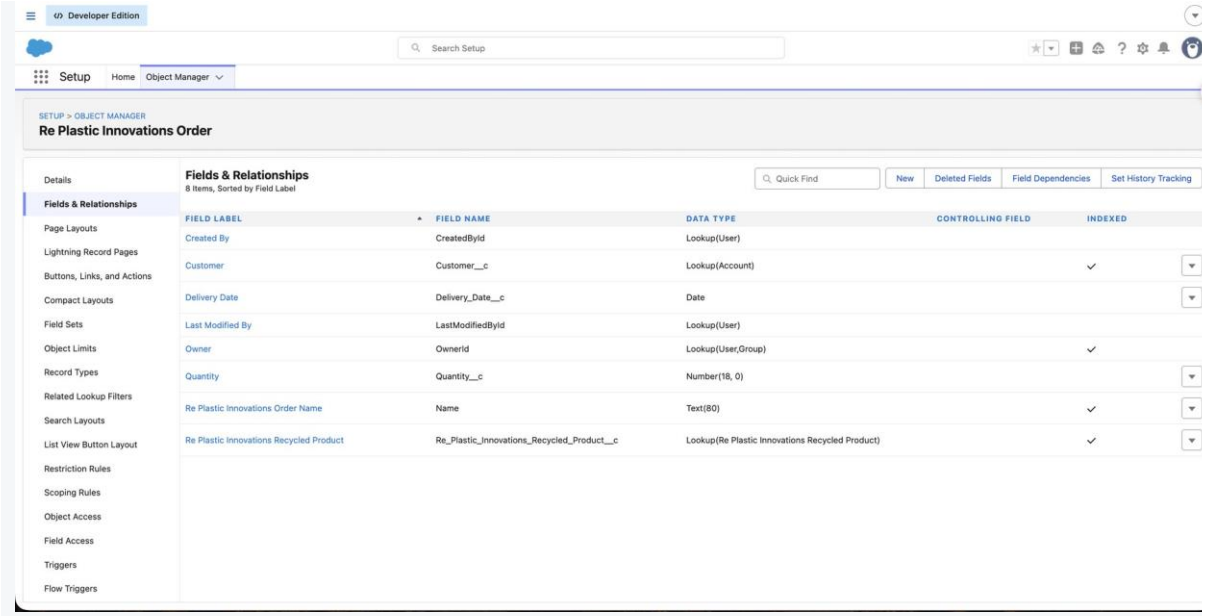


Figure 3: Re Plastic Innovations Order: fields and relationships (Customer, Delivery Date, Quantity, Recycled Product)

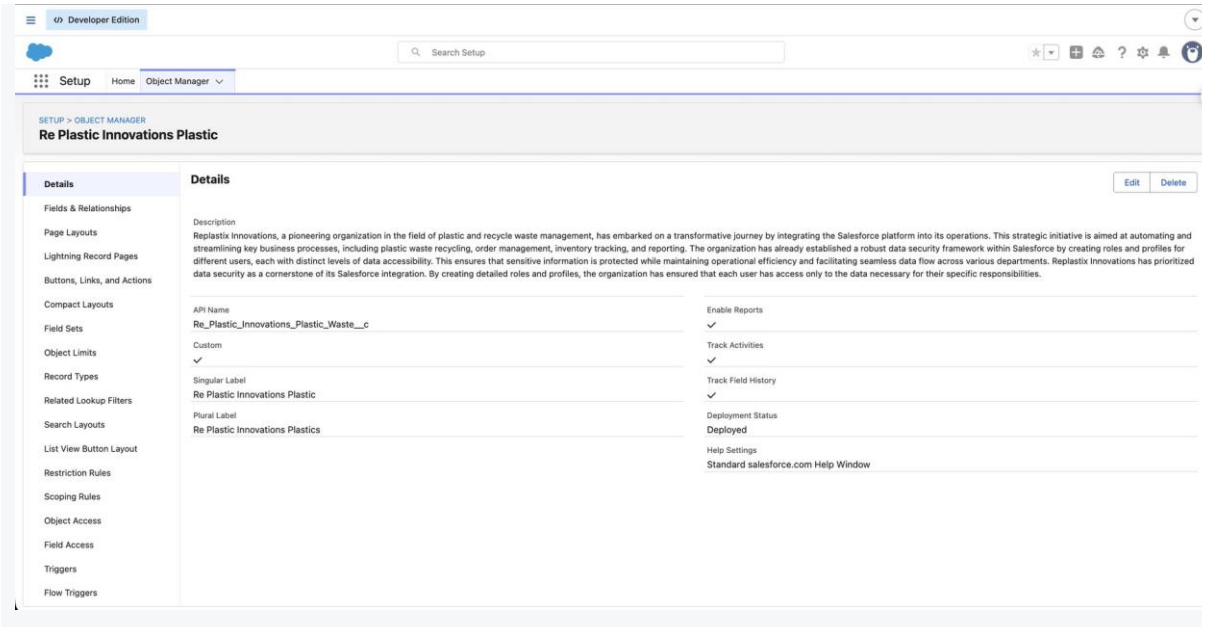


Figure 4: Re Plastic Innovations Plastic: object details (API name Re_Plastic_Innovations_Plastic_Waste__c)

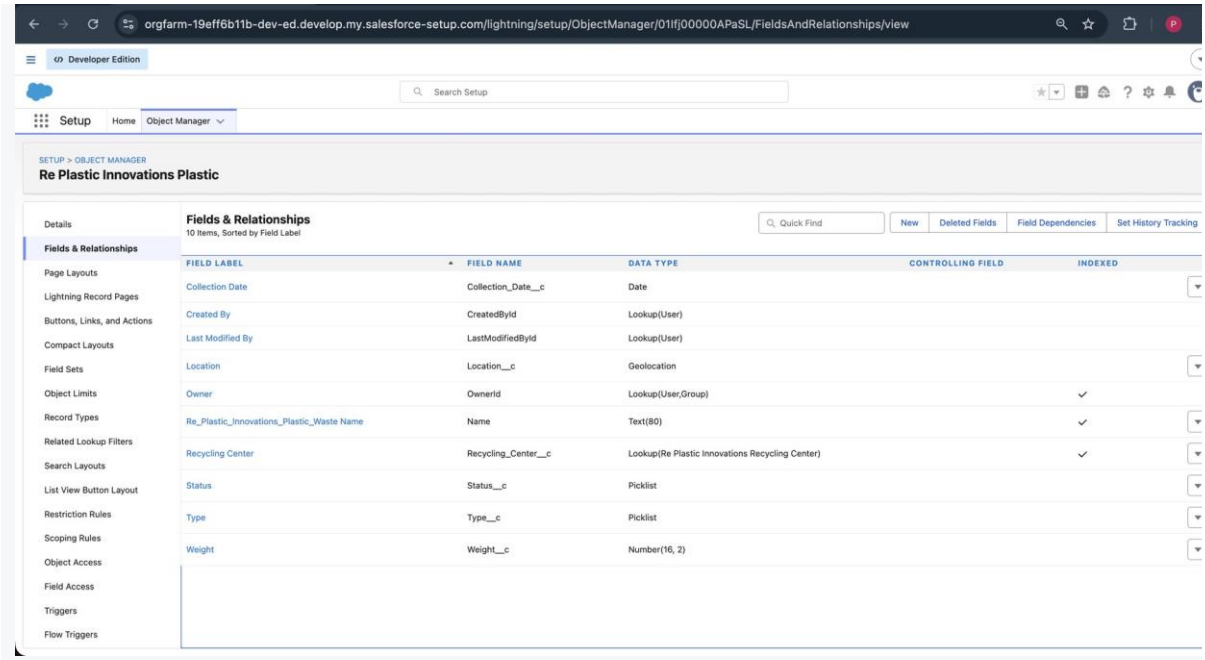


Figure 5: Re Plastic Innovations Plastic: fields and relationships (Collection Date, Location, Recycling Center, Status, Type, Weight)

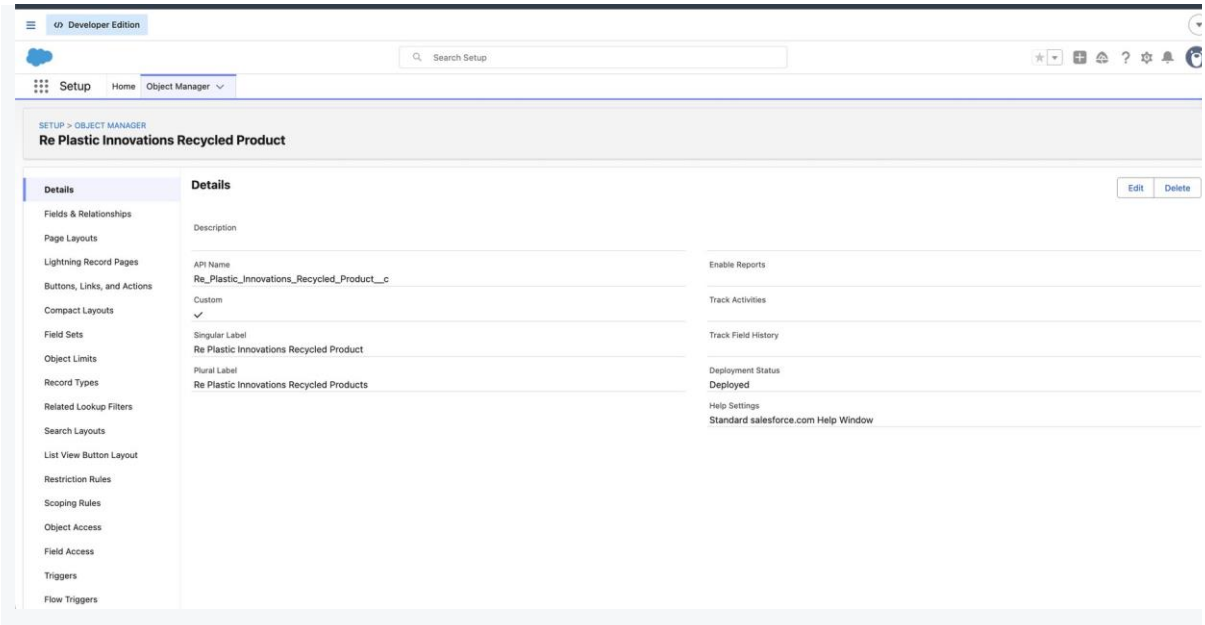


Figure 6: Re Plastic Innovations Recycled Product: object details

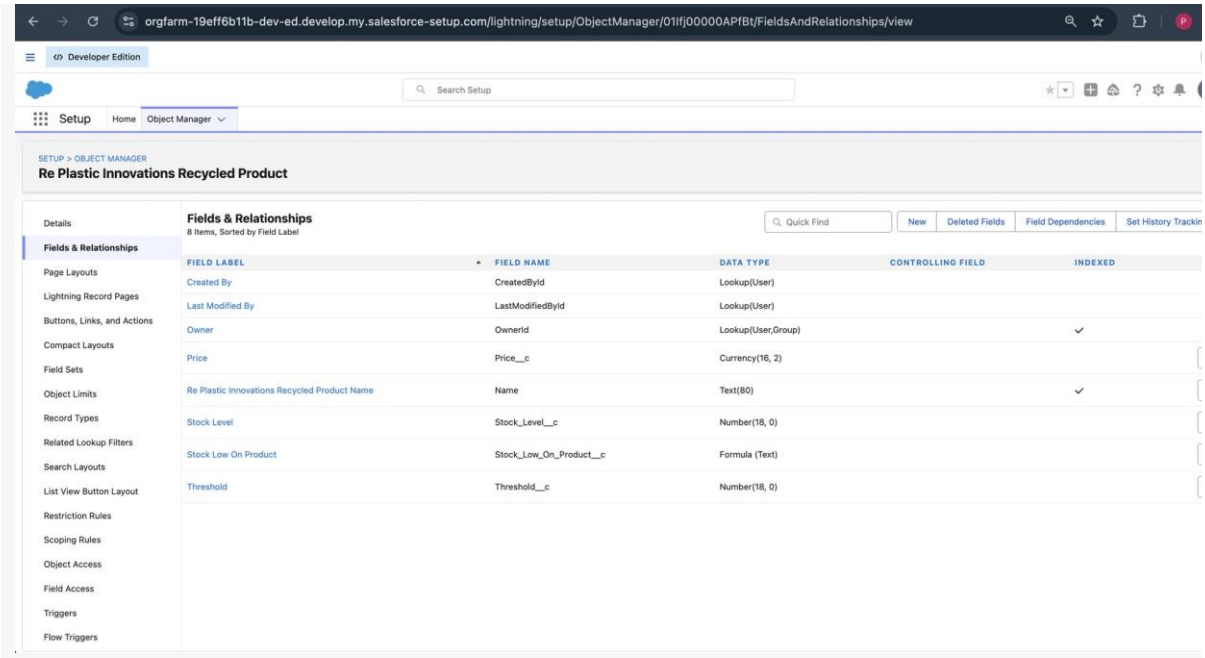


Figure 7: Re Plastic Innovations Recycled Product: fields (Price, Stock Level, Stock Low On Product formula, Threshold)

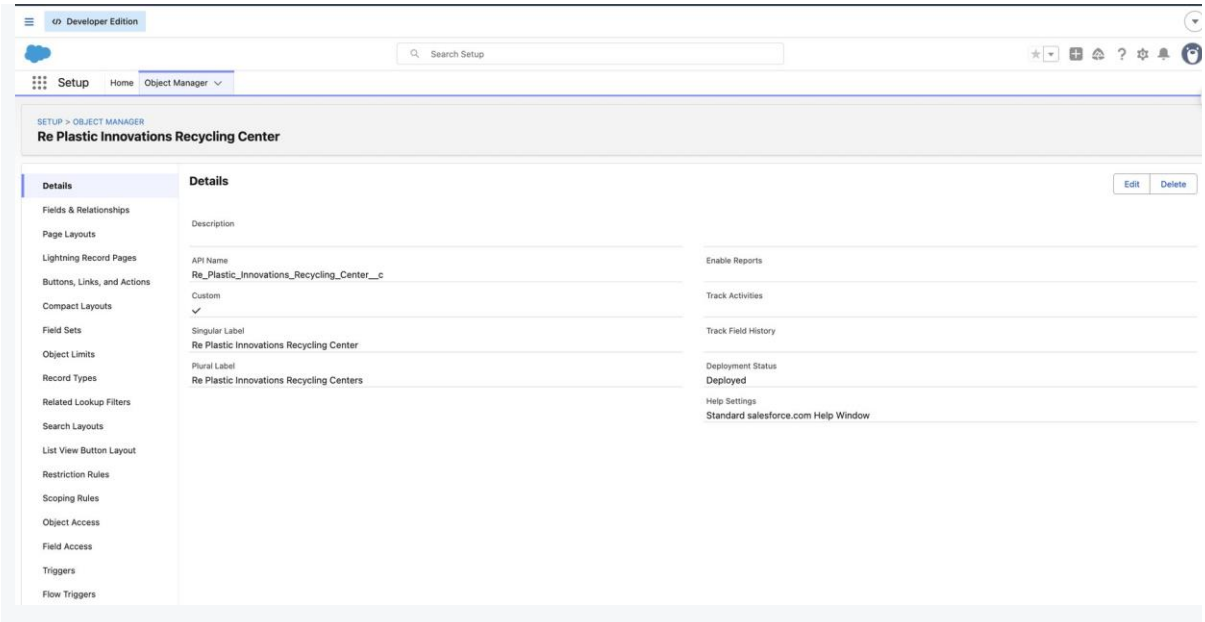


Figure 8: Re Plastic Innovations Recycling Center: object details

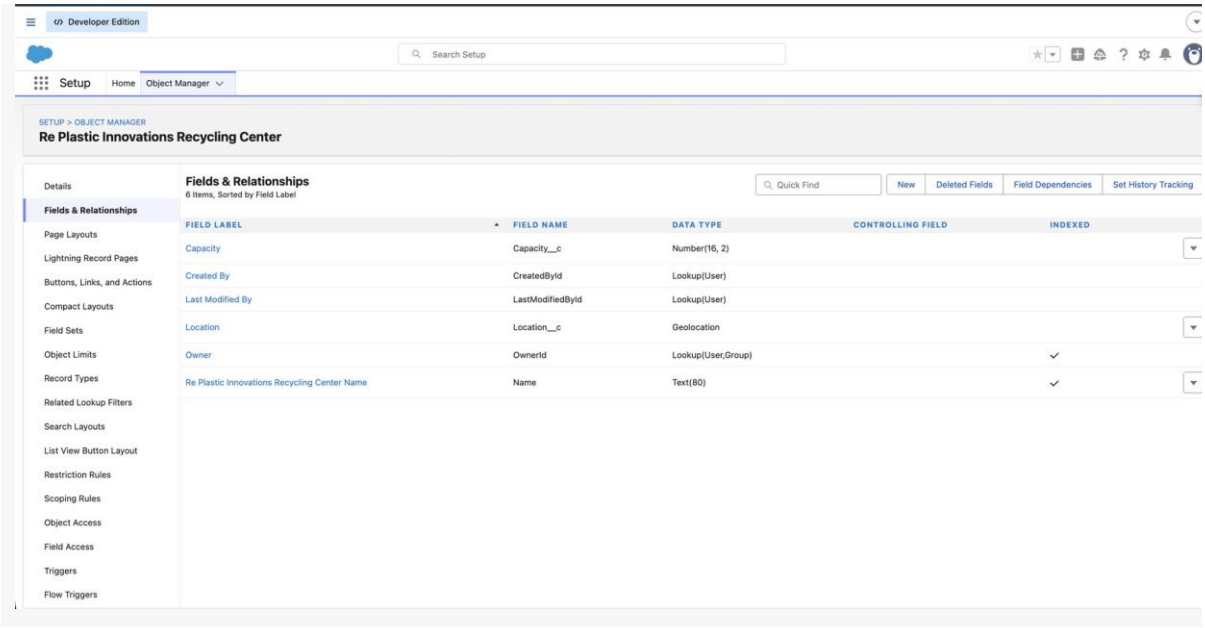


Figure 9: Re Plastic Innovations Recycling Center: fields (Capacity, Location)

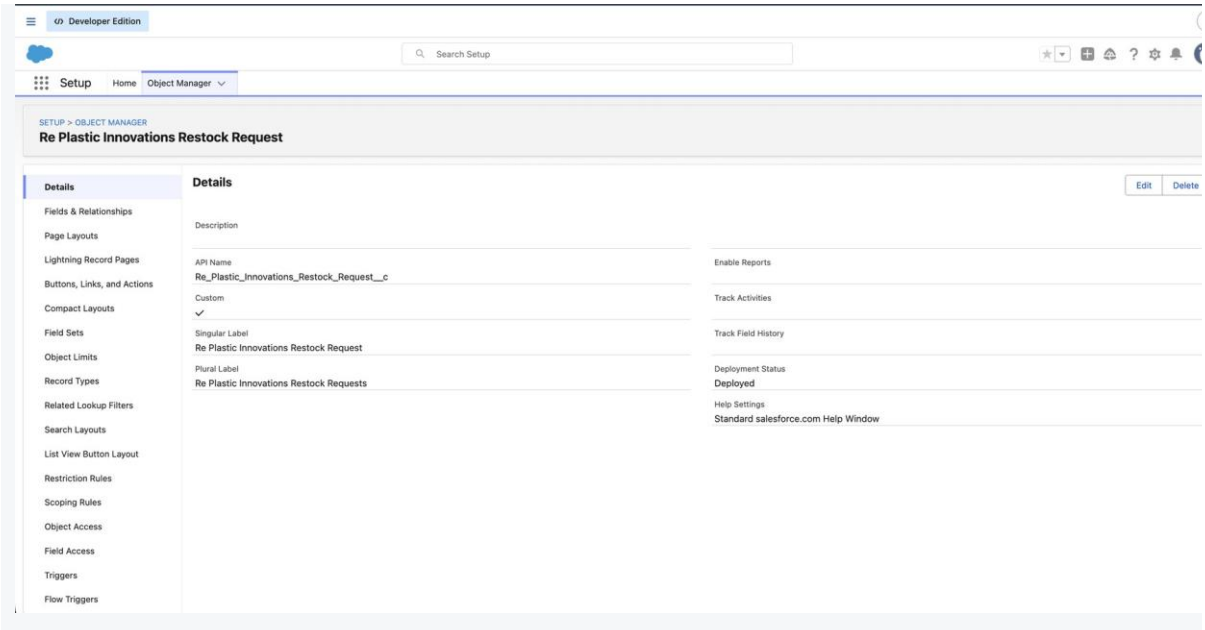


Figure 10: Re Plastic Innovations Restock Request: object details

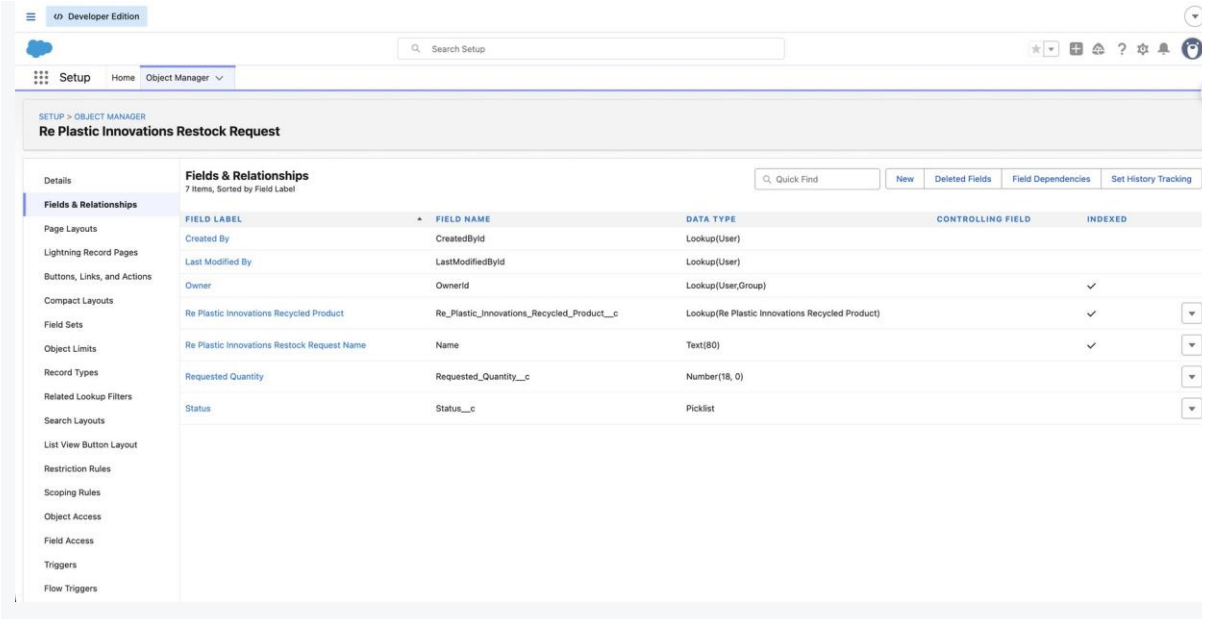


Figure 11: Re Plastic Innovations Restock Request: fields (Recycled Product, Requested Quantity, Status)

2.2 Automation Logic

The project uses declarative and programmatic automation according to the type of requirement. A scheduled Flow handles the time-based low-stock check, while Apex triggers handle transaction-based inventory processing that requires related-record queries, calculations, and updates.

Automation	When it runs	Main action
Scheduled Flow — Stock Level Is Low	Daily at 6:00 AM	Checks Stock Level against Threshold and creates a High-priority Task when stock is low
UpdateStockAfterOrder	After Order insert	Calls InventoryManager to reduce stock or create a shortage restock request
UpdateStockAfterRestockApproval	After Restock Request update	Detects Pending → Approved, increases stock, and calls email notification helper

The scheduled Flow "Stock Level Is Low – V1" is active and starts daily at 6:00 AM. After the start element, Decision 1 evaluates the stock condition. Its first outcome leads to a Create Task element that creates a Task with Priority set to High and Assigned To ID populated from the product record; the Default Outcome ends the Flow without action.

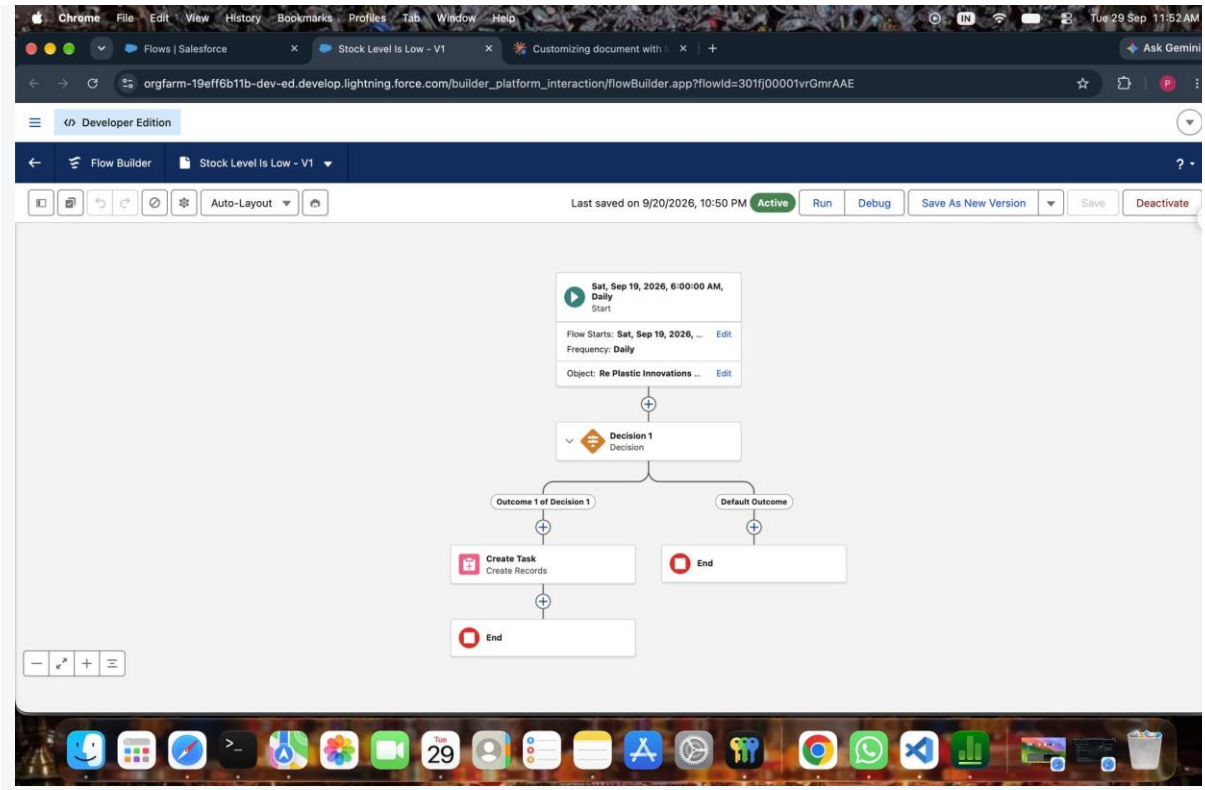


Figure 12: Scheduled Flow "Stock Level Is Low – V1": daily 6:00 AM start, Decision 1, and Create Task

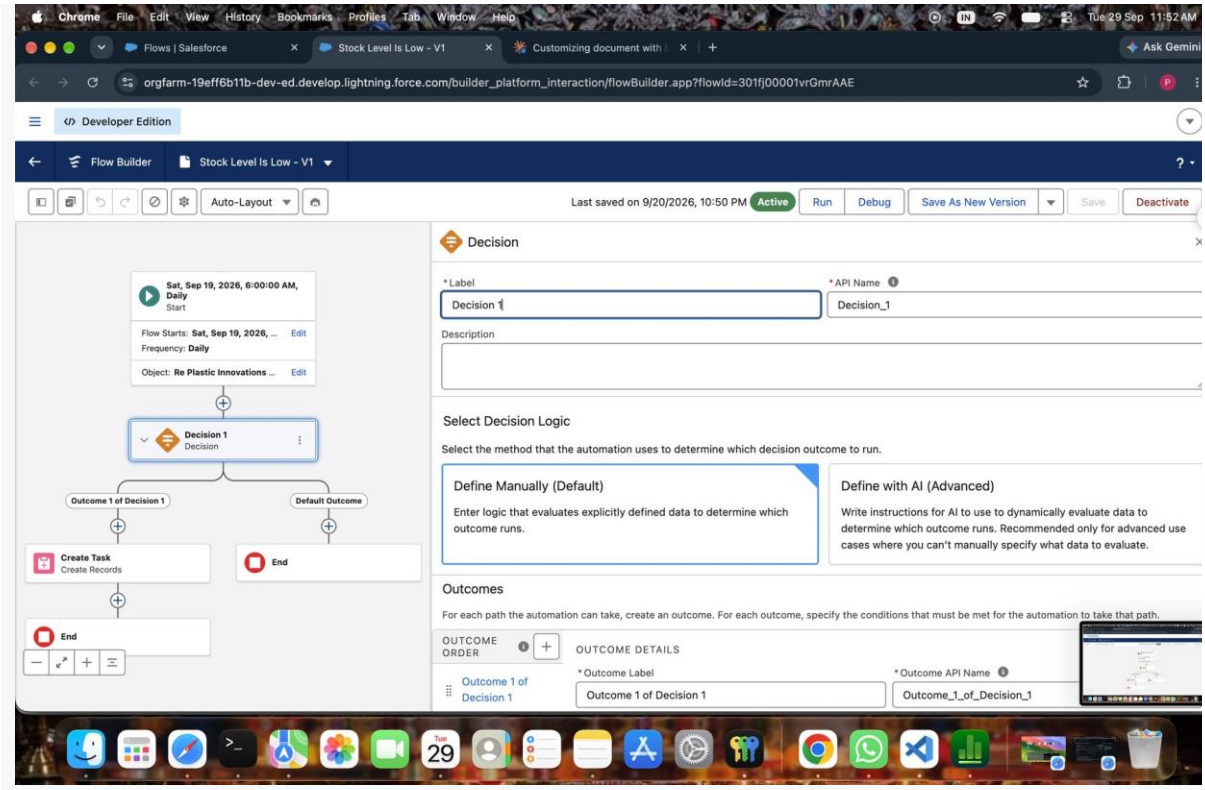


Figure 13: Flow Builder: Decision 1 configuration and outcomes

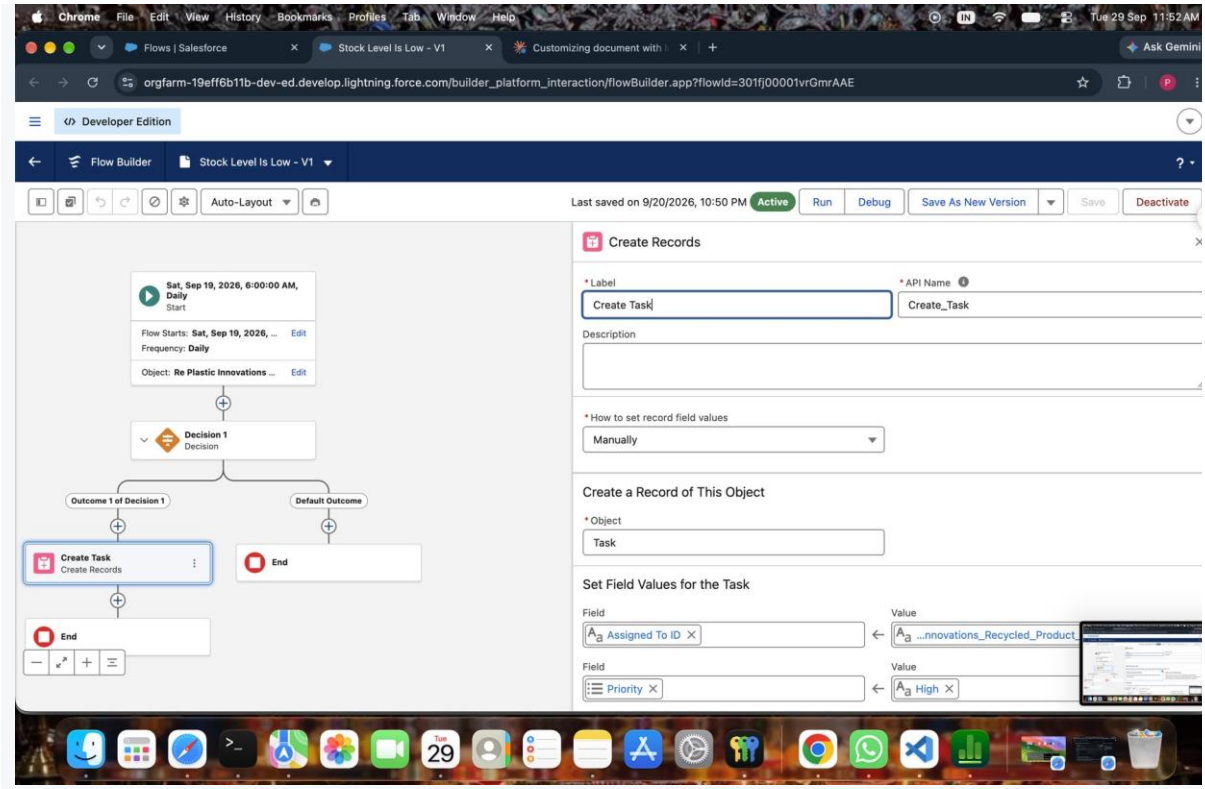


Figure 14: Flow Builder: Create Task element (Assigned To ID and Priority = High)

The Apex layer has four components: the InventoryManager class, which processes order stock and restock approvals; the EmailNotificationHelper class, whose sendRestockNotification method emails the current user; the UpdateStockAfterOrder trigger (after insert on Order), which passes Trigger.new to InventoryManager.processOrderStock; and the UpdateStockAfterRestockApproval trigger (after update on Restock Request), which collects records whose Status changed to Approved and passes them to InventoryManager.processRestockApproval.

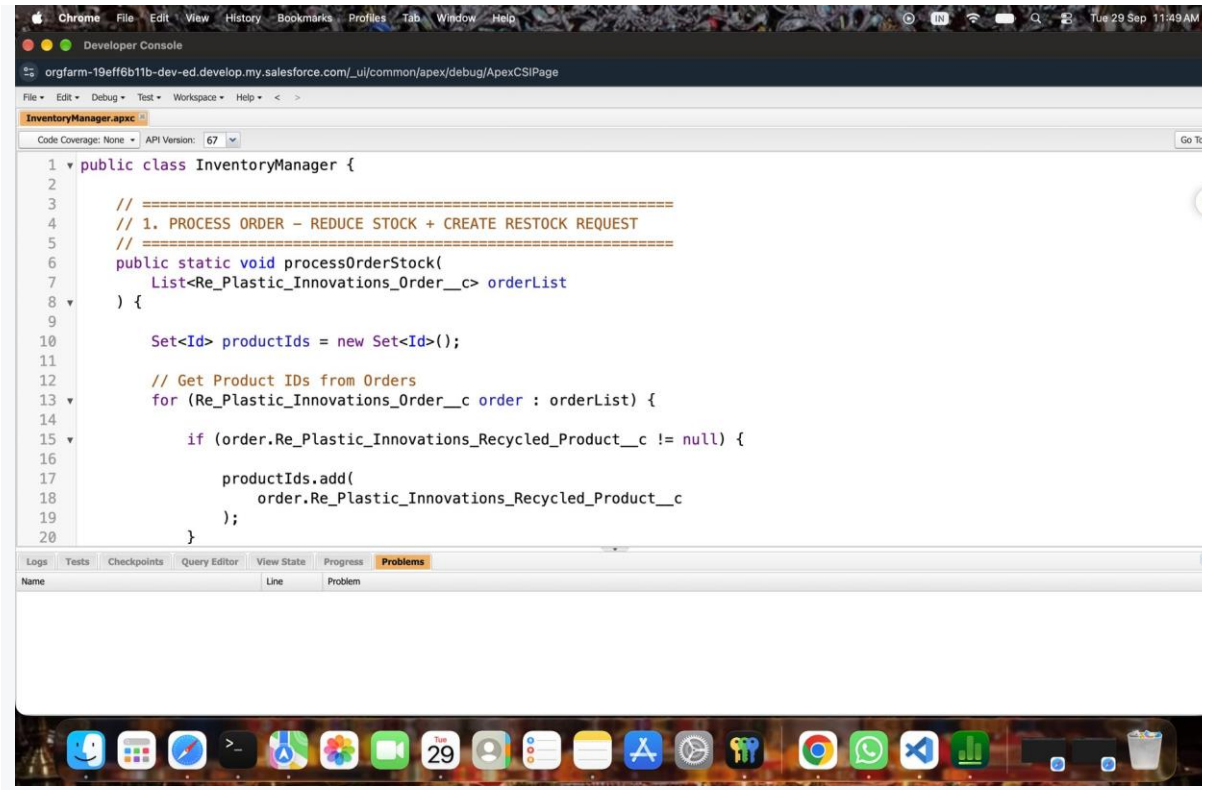


Figure 15: Apex class InventoryManager (processOrderStock)

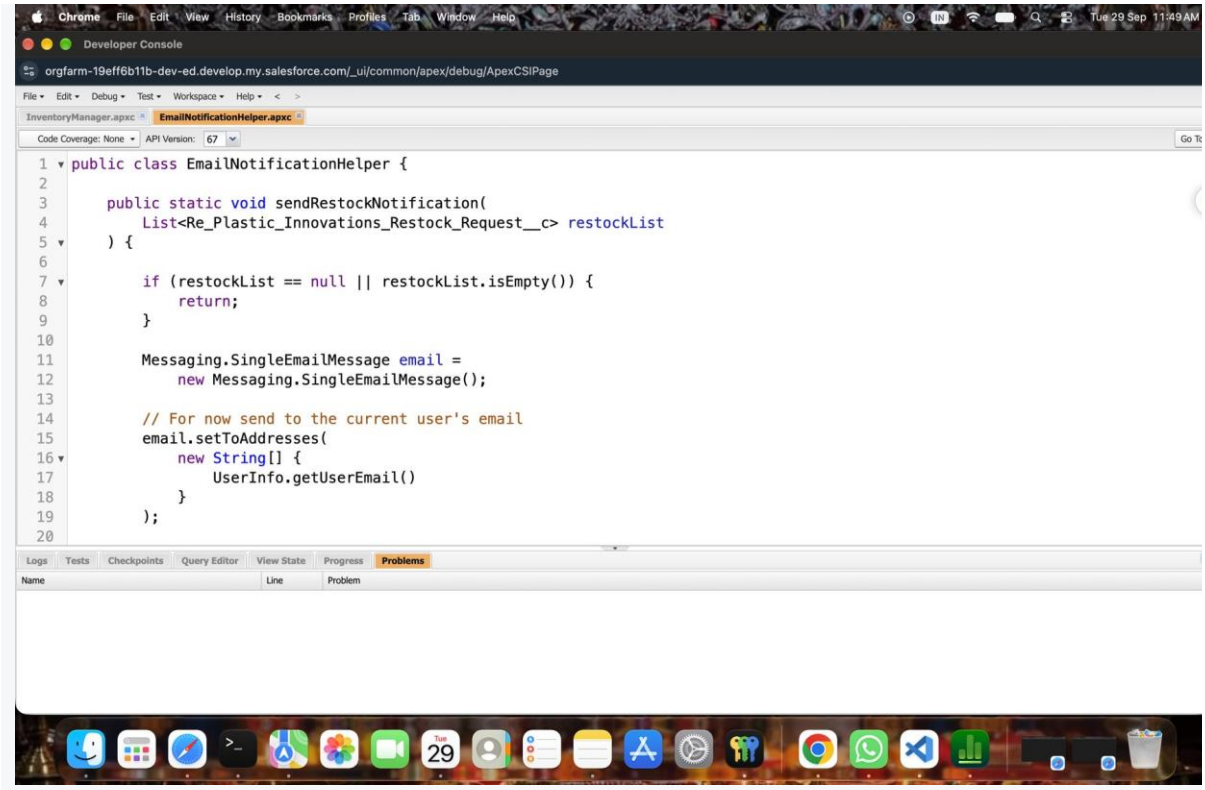


Figure 16: Apex class EmailNotificationHelper (sendRestockNotification)

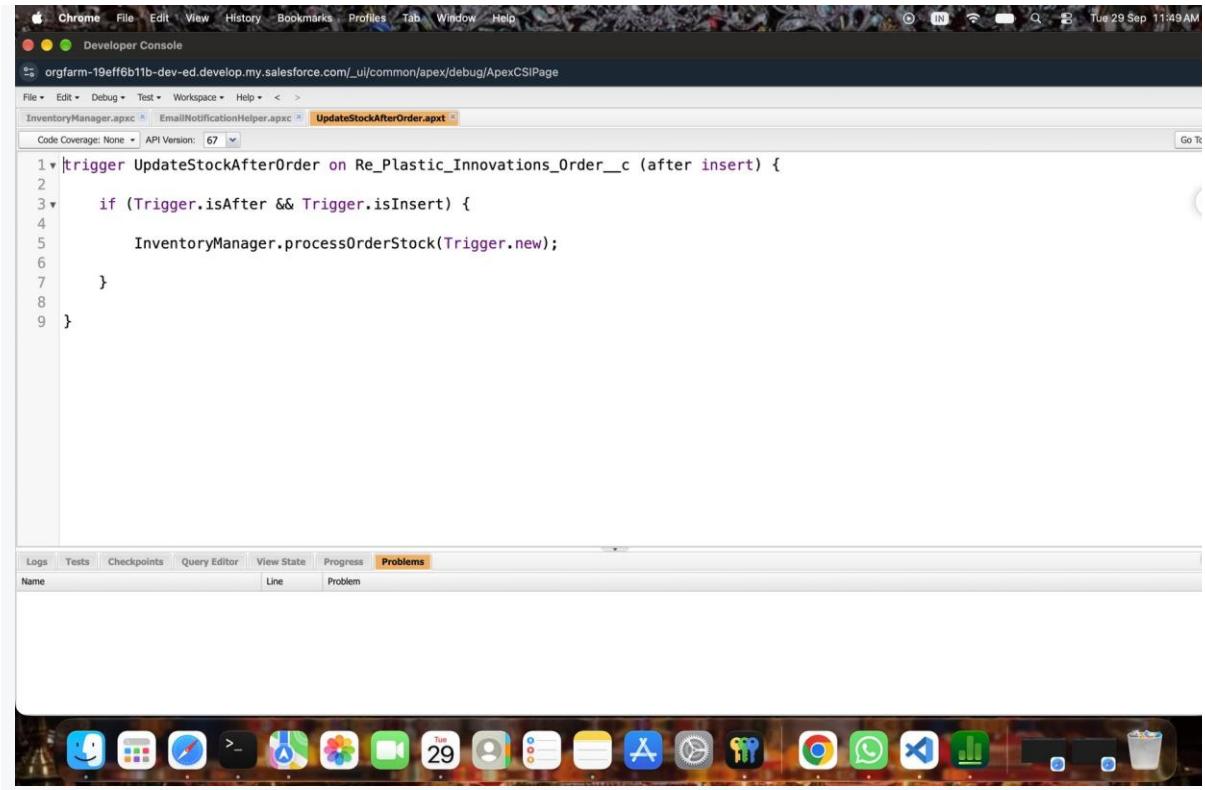


Figure 17: Trigger UpdateStockAfterOrder on Re Plastic Innovations Order (after insert)

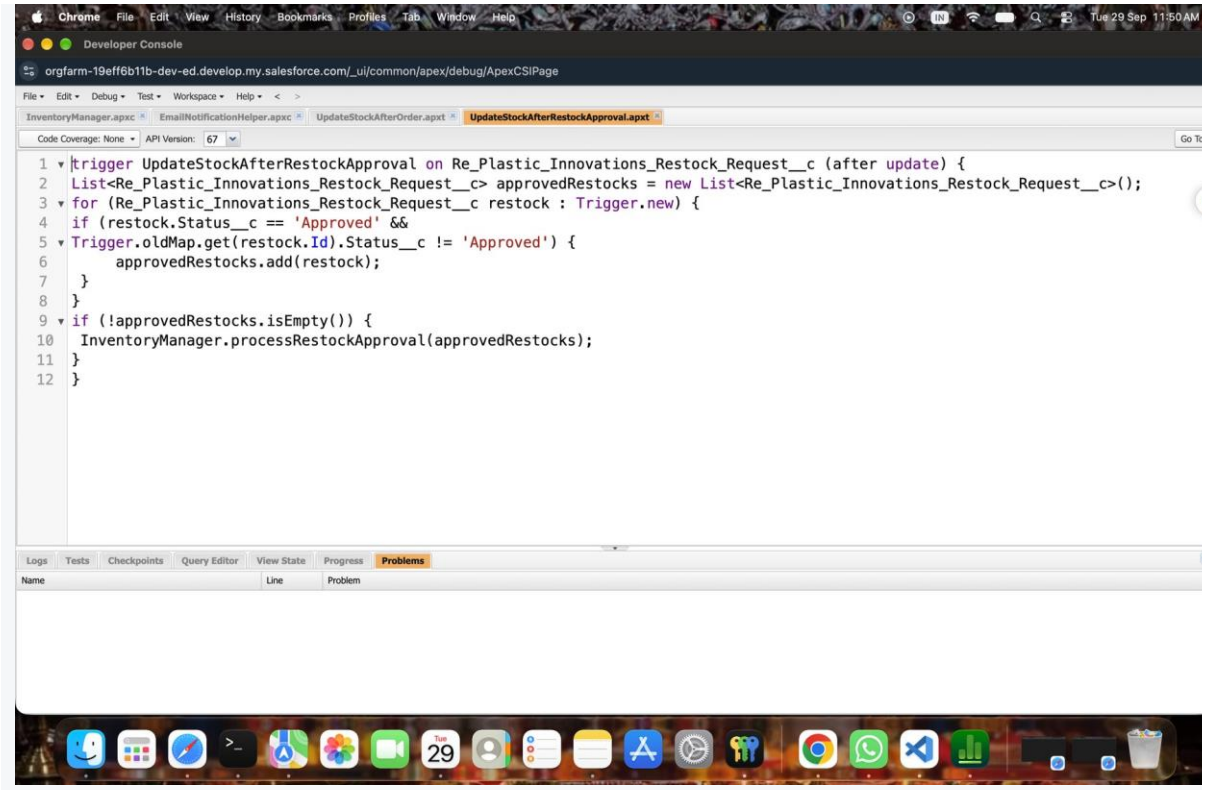


Figure 18: Trigger UpdateStockAfterRestockApproval on Re Plastic Innovations Restock Request (after update)

2.3 Business Rules

Business rules are implemented through formulas, validation rules, Flow decisions, and Apex logic. The design keeps simple field-level rules declarative and uses Apex for multi-record transaction processing.

- The Stock Low On Product formula field displays 'Low Stock - Restock Needed' when Stock Level is below Threshold; otherwise it displays 'Sufficient Stock'.
- Order quantity must be greater than zero.
- Plastic Waste collection date cannot be in the future.
- When sufficient stock exists, an order reduces the related product stock by the ordered quantity.
- When stock is insufficient, the system creates a Pending Restock Request for the shortage amount.
- When a restock request changes to Approved, the product stock is increased by the requested quantity and the notification helper is invoked.

PHASE 3 — UI/UX DEVELOPMENT & CUSTOMIZATION

3.1 Interface Design

A dedicated Lightning application named Re Plastic Innovations was created as the central workspace. Custom tabs were created for the project objects so that the main business records are directly accessible from the application navigation. The navigation bar shows the Re Plastic Innovations Orders, Plastics, Recycled Products, and Restock Requests tabs, with additional tabs available under More.

The interface is organized around the operational workflow rather than around technical configuration. Users can move between waste records, recycling centers, recycled products, orders, and restock requests without navigating through Salesforce Setup.

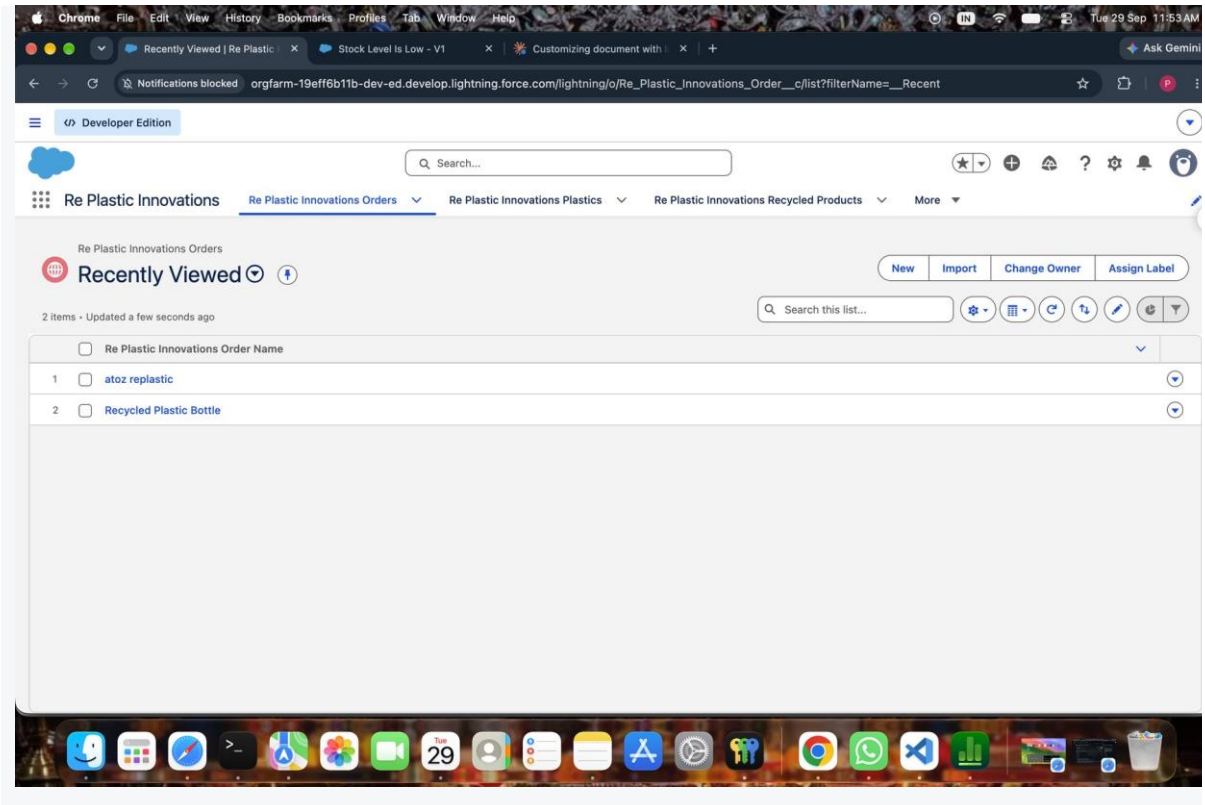


Figure 19: Re Plastic Innovations Lightning app: Orders tab and list view

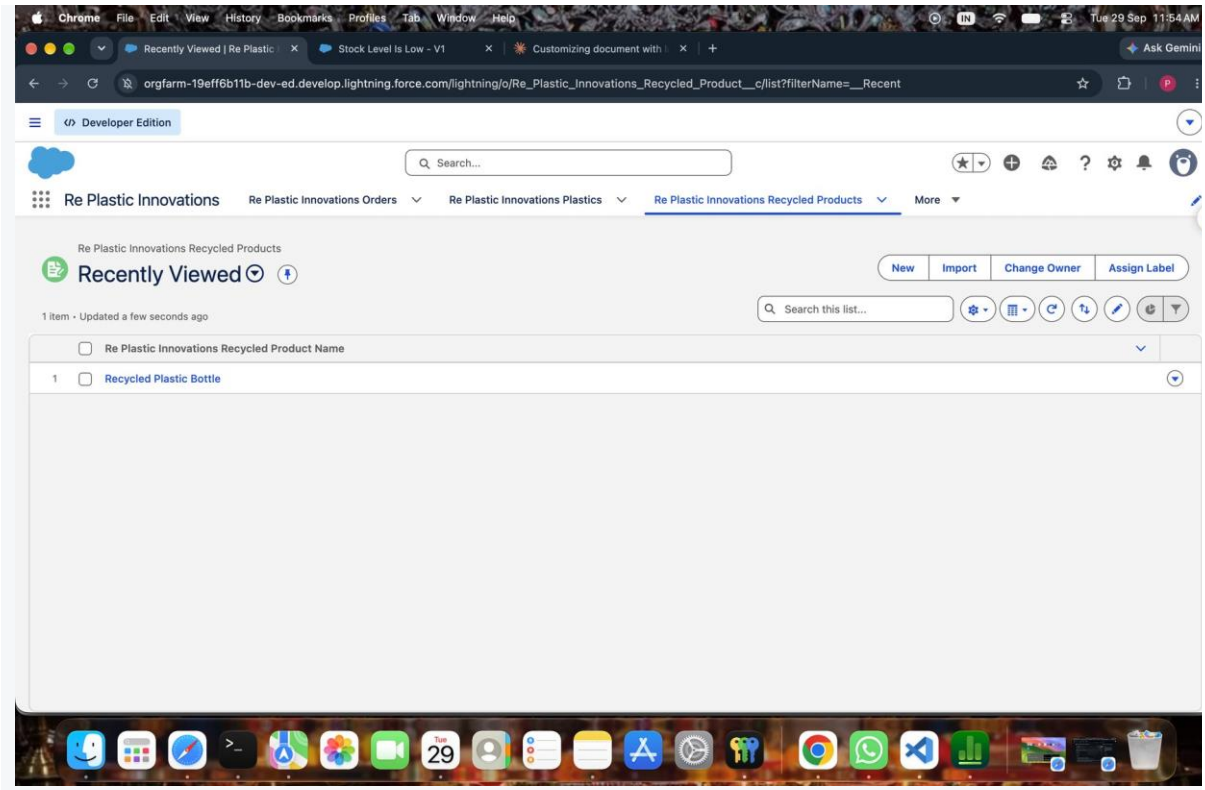


Figure 20: Re Plastic Innovations Recycled Products tab

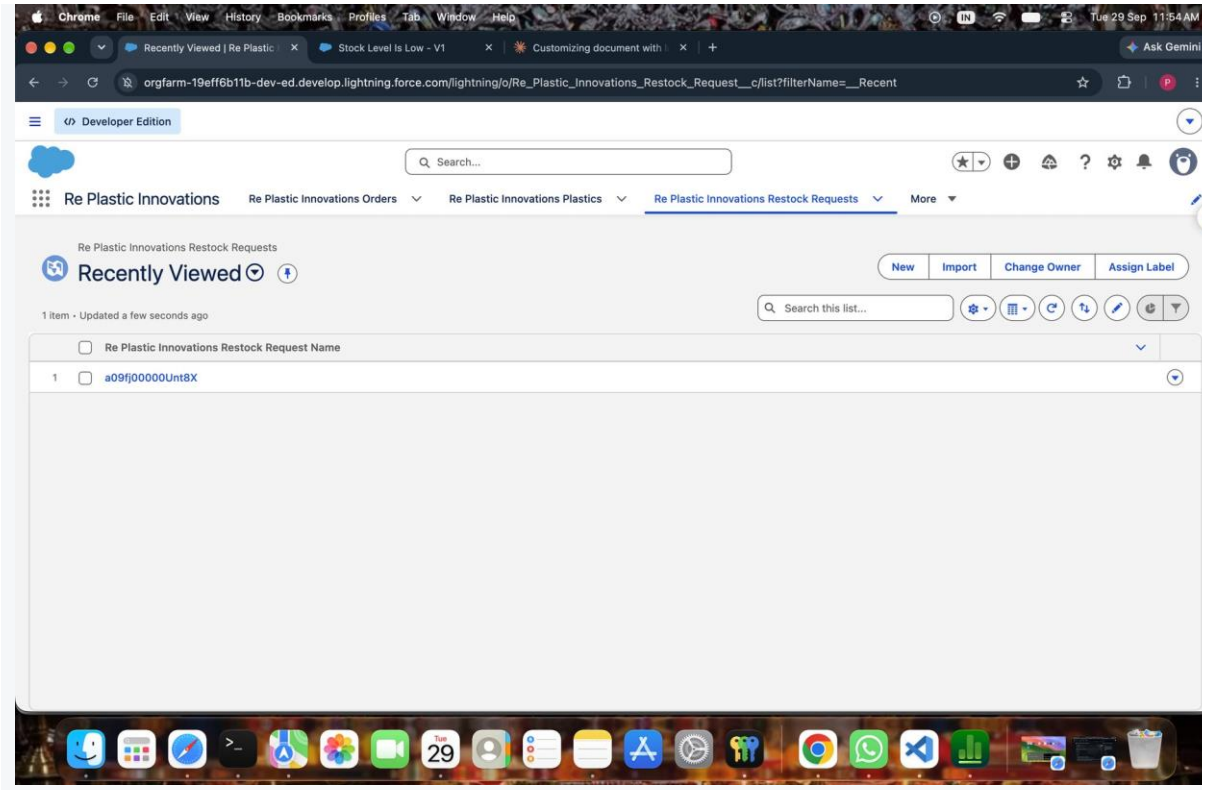


Figure 21: Re Plastic Innovations Restock Requests tab

3.2 Navigation Flow

The navigation flow follows the business process. Waste records can be associated with recycling centers, recycled products hold the resulting inventory, customer orders consume available stock, and restock requests handle shortages.

Starting Point	Next Record / Action	Purpose
Plastic Waste	Recycling Center	Associate collected waste with a processing location
Recycled Product	Order	Process customer demand against product stock
Order	Restock Request	Handle a shortage when available stock is insufficient
Restock Request	Recycled Product	Increase stock after the request is approved

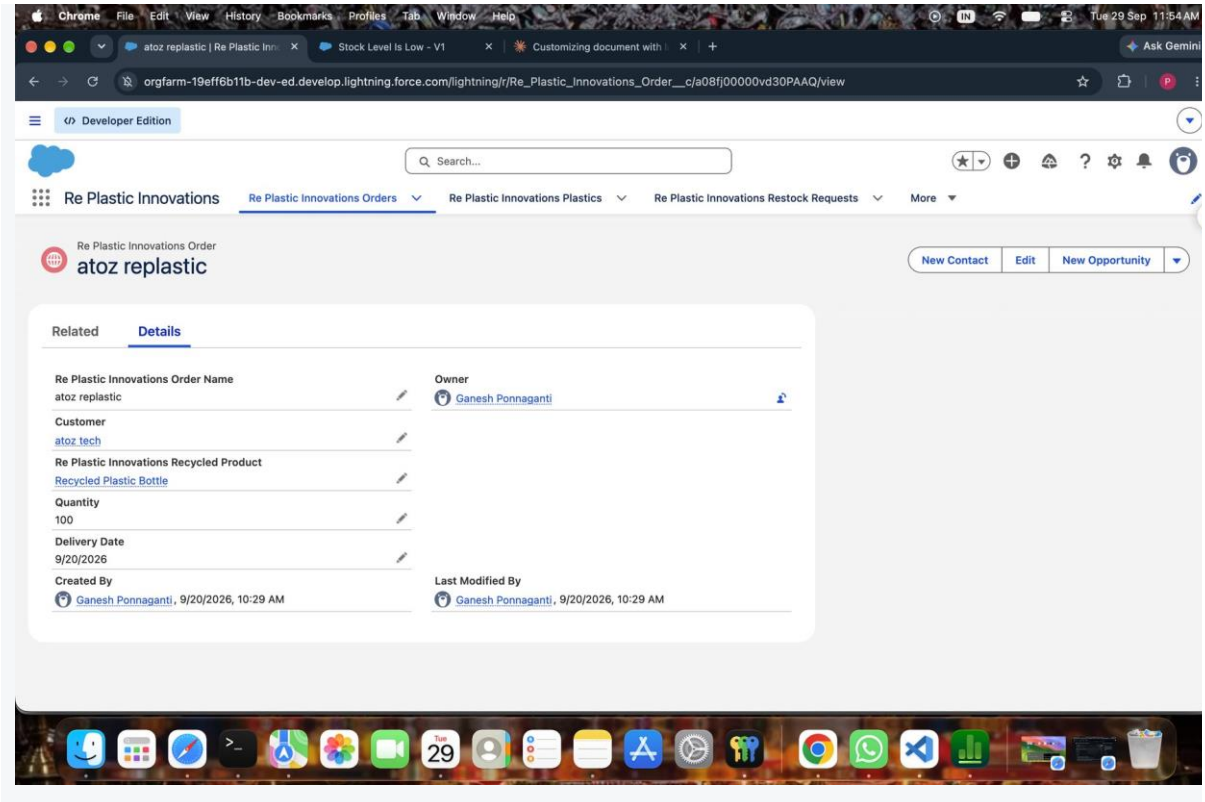


Figure 22: Order record detail page showing the Customer and Recycled Product lookups

3.3 Usability

Usability was addressed through clear object names, direct navigation tabs, meaningful picklist values, validation messages, and automatic status calculations. The system provides immediate feedback when a user enters an invalid quantity or future collection date.

The stock-status formula also makes inventory conditions visible without requiring users to manually compare two fields. Automation reduces repetitive checking, while the role-based design limits users to the information relevant to their responsibilities.

PHASE 4 — DATA MIGRATION, TESTING & SECURITY

4.1 Data Handling

The project data model was created with dedicated fields for operational information such as weight, type, dates, quantities, stock levels, thresholds, locations, and statuses. Lookup relationships were used to maintain links between related records.

Each object uses the standard Name field (Text, 80). Order and Recycled Product records use descriptive names such as "Recycled Plastic Bottle", while Restock Request records are identified by their record ID.

Sample records used to demonstrate the workflow: the Order "atoz replastic" (customer atoz tech, quantity 100, delivery date 9/20/2026), the Order "Recycled Plastic Bottle" (customer Nadhu enterprises, quantity 60, delivery date 9/22/2026), and a Restock Request for 500 units of Recycled Plastic Bottle with status Approved.

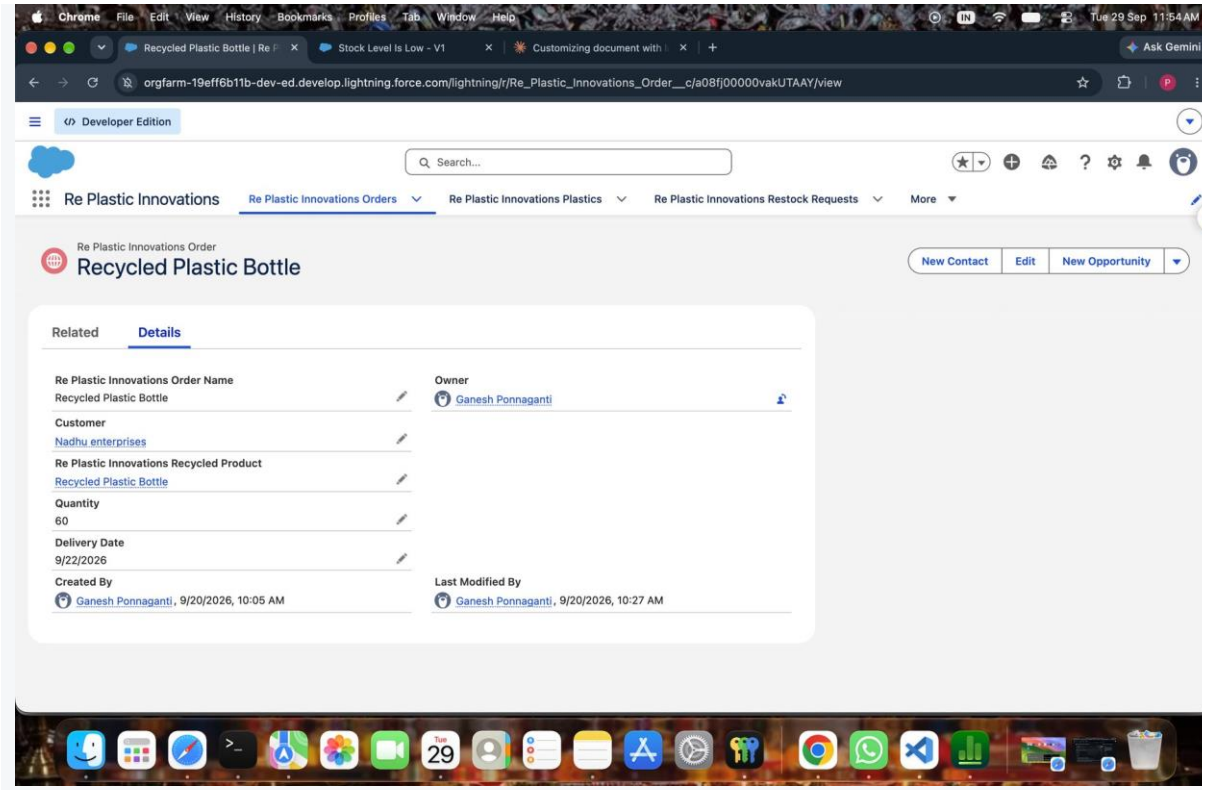


Figure 23: Order record "Recycled Plastic Bottle": customer Nadhu enterprises, quantity 60, delivery date 9/22/2026

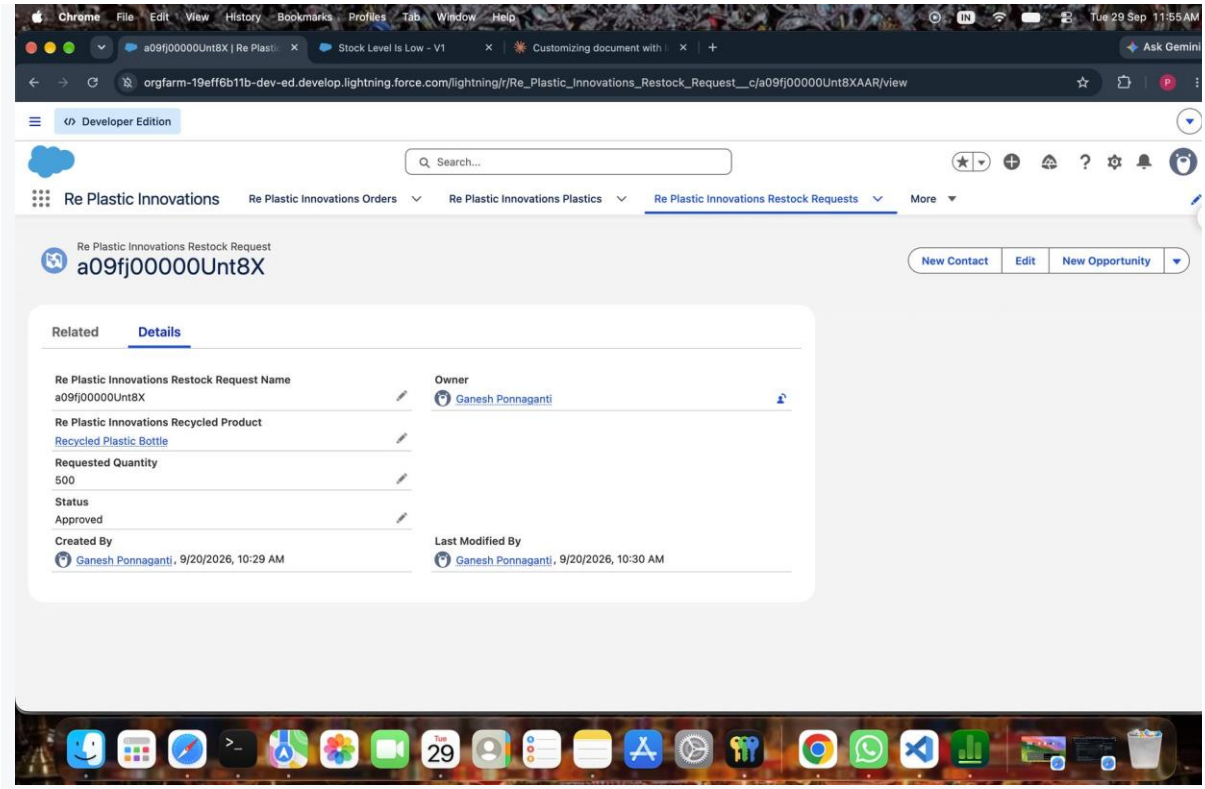


Figure 24: Restock Request record: Recycled Plastic Bottle, requested quantity 500, status Approved

4.2 Access Control

Access control was implemented at both object and record levels. Profiles define what users can do with objects, roles establish the organizational hierarchy, private organization-wide defaults restrict record visibility, and sharing rules provide the required additional access.

Control	Purpose	Implemented Example
Profiles	Object-level permissions	Platform 1, Platform 2, Platform 3
Roles	Hierarchy and sharing context	CEO → Recycling Manager / Sales Representative → Warehouse Supervisor
OWD	Default record visibility	Relevant custom objects configured as Private
Sharing Rules	Additional record access	CEO to Recycling Manager; CEO to Sales Representative; Sales Representative to Warehouse Supervisor

4.3 QA Testing

The Apex test class covers the core inventory scenarios required by the project. Test data starts with a recycled product having stock of 50 and threshold of 10. The tests then verify sufficient-stock order processing, insufficient-stock restock creation, and stock increase after restock approval.

Test Scenario	Expected Result	Observed Result
Order quantity within available stock	Stock decreases by ordered quantity	Passed
Order quantity exceeds available stock	Pending restock request is created for shortage	Passed
Restock request changes to Approved	Product stock increases by requested quantity	Passed

4.4 Data Integrity

Data integrity is supported by validation rules, formula logic, controlled picklist values, and Apex processing. Invalid order quantities are rejected, future collection dates are blocked, and stock changes are performed by the implemented business logic instead of relying on manual calculation.

The restock trigger also checks that the status has newly changed to Approved before processing, preventing the same approval event from being repeatedly processed by later edits.

PHASE 5 — DEPLOYMENT, DOCUMENTATION & FINAL PRESENTATION

5.1 Troubleshooting

Troubleshooting focused on verifying field API names, lookup relationships, Flow references, trigger execution, stock calculations, email configuration, and test behavior. One important implementation detail was ensuring that the scheduled Flow assigned the generated Task to the product owner through the Assigned To ID field, with Priority set to High.

Apex execution was verified using the project test scenarios. The order trigger and restock-approval trigger were kept focused on their respective business events so that each automation has a clear responsibility.

5.2 Adherence to Timelines

The work was organized progressively: configuration and data structure were completed first, followed by automation, Apex logic, security, and testing. This sequence allowed each later stage to use the components established in the earlier stages and reduced rework.

5.3 Innovation

The project combines Salesforce declarative capabilities with Apex processing to create a practical inventory workflow for recycled products. Instead of treating stock monitoring, shortage creation, restocking, and notification as separate manual activities, the implementation connects them into a single business process.

5.4 Document Functional Overview

Functionally, the system supports five connected areas: plastic-waste collection, recycling-center management, recycled-product inventory, customer orders, and restock requests. Users can maintain records through the Lightning application while automation handles repetitive monitoring and transaction processing.

The main operational scenario is: a customer order is created → available stock is checked → stock is reduced when sufficient → otherwise a Pending Restock Request is created → the request can be changed to Approved → product stock is increased and an email notification is sent.

5.5 Document Technical Blueprint

The technical design consists of custom Salesforce objects and lookup relationships at the data layer; a Lightning application and custom tabs at the interface layer; validation rules and a formula at the business-rule layer; Scheduled Flow for daily low-stock monitoring; and Apex classes/triggers for transaction-based inventory processing.

Layer	Implemented Components
Data	Plastic Waste, Recycling Center, Recycled Product, Order, Restock Request, Account lookup
UI	Re Plastic Innovations Lightning App and custom tabs
Declarative Logic	Stock status formula, two validation rules, scheduled low-stock Flow
Programmatic Logic	InventoryManager, two Apex triggers, EmailNotificationHelper
Security	Profiles, roles, private OWD, sharing rules
Testing	InventoryManagerTest covering three core scenarios

5.6 Document Setup Manual

The setup sequence for reproducing the implementation is:

- Create the five custom objects and required fields.
- Create lookup relationships and custom tabs.
- Create the Re Plastic Innovations Lightning application and add the required navigation items.
- Configure the stock-status formula and validation rules.
- Create the scheduled low-stock Flow and activate it.
- Create InventoryManager, the order trigger, the restock-approval trigger, and EmailNotificationHelper.
- Configure profiles, roles, private organization-wide defaults, and sharing rules.
- Create representative records and run the InventoryManagerTest class.

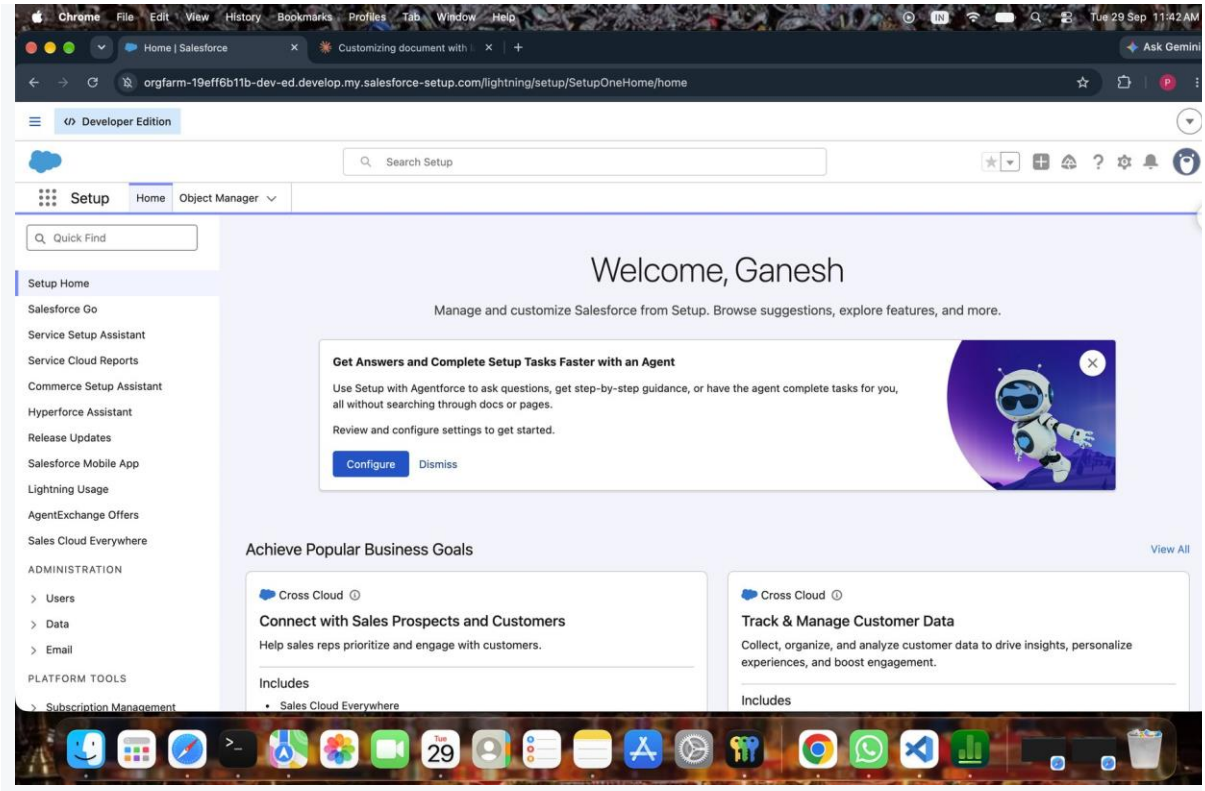


Figure 25: Salesforce Developer Edition: Setup Home used to reach Object Manager, Flows, and the Developer Console

5.7 Project Demo Video Planning

The demonstration video should be kept between two and five minutes and should focus on the working business scenario rather than showing every configuration screen.

- Introduce the student and project.
- State the problem and the Salesforce-based solution.
- Show the Re Plastic Innovations application and main navigation.
- Demonstrate a sufficient-stock order and show the stock reduction.
- Demonstrate an insufficient-stock order and show the automatically created Pending Restock Request.
- Change the request to Approved and show the stock increase and notification behavior.
- Briefly explain why Flow was used for scheduled low-stock monitoring and Apex triggers were used for transaction-based inventory processing.
- Finish by showing successful test results and summarizing the outcome.

5.8 Visual Demonstration

The final presentation should use screenshots that prove the important implemented features. The strongest evidence is a combination of application navigation, product inventory, automation, the shortage/restock workflow, security configuration, and successful testing.

Figure	Screenshot Included
1	1.2 — Object Manager showing the five Re Plastic Innovations custom objects
2	2.1 — Re Plastic Innovations Order: object details (API name Re_Plastic_Innovations_Order__c)
3	2.1 — Re Plastic Innovations Order: fields and relationships (Customer, Delivery Date, Quantity, Recycled Product)
4	2.1 — Re Plastic Innovations Plastic: object details (API name Re_Plastic_Innovations_Plastic_Waste__c)
5	2.1 — Re Plastic Innovations Plastic: fields and relationships (Collection Date, Location, Recycling Center, Status, Type, Weight)
6	2.1 — Re Plastic Innovations Recycled Product: object details
7	2.1 — Re Plastic Innovations Recycled Product: fields (Price, Stock Level, Stock Low On Product formula, Threshold)
8	2.1 — Re Plastic Innovations Recycling Center: object details
9	2.1 — Re Plastic Innovations Recycling Center: fields (Capacity, Location)
10	2.1 — Re Plastic Innovations Restock Request: object details
11	2.1 — Re Plastic Innovations Restock Request: fields (Recycled Product, Requested Quantity, Status)
12	2.2 — Scheduled Flow "Stock Level Is Low – V1": daily 6:00 AM start, Decision 1, and Create Task
13	2.2 — Flow Builder: Decision 1 configuration and outcomes
14	2.2 — Flow Builder: Create Task element (Assigned To ID and Priority = High)
15	2.2 — Apex class InventoryManager (processOrderStock)
16	2.2 — Apex class EmailNotificationHelper (sendRestockNotification)
17	2.2 — Trigger UpdateStockAfterOrder on Re Plastic Innovations Order (after insert)
18	2.2 — Trigger UpdateStockAfterRestockApproval on Re Plastic Innovations Restock Request (after update)
19	3.1 — Re Plastic Innovations Lightning app: Orders tab and list view
20	3.1 — Re Plastic Innovations Recycled Products tab
21	3.1 — Re Plastic Innovations Restock Requests tab
22	3.2 — Order record detail page showing the Customer and Recycled Product lookups
23	4.1 — Order record "Recycled Plastic Bottle": customer Nadhu enterprises, quantity 60, delivery date 9/22/2026
24	4.1 — Restock Request record: Recycled Plastic Bottle, requested quantity 500, status Approved
25	5.6 — Salesforce Developer Edition: Setup Home used to reach Object Manager, Flows, and the Developer Console

5.9 Explanation Clarity

The project should be explained using a simple sequence: problem, solution, data model, automation, live scenario, technical choices, testing, and result. Technical terms should be connected to their purpose. For example, validation rules are used for simple data restrictions, Scheduled Flow is used for time-based low-stock checking, and Apex is used where the process requires custom logic across related records.

The restock process should be described accurately as a status-based workflow: a Pending request is created automatically, a user changes it to Approved, and the after-update trigger processes the approval. No Salesforce Approval Process should be claimed because that configuration was not implemented.

5.10 Realism & Quality

The solution represents a realistic operational scenario in which recycled-product demand affects available stock and shortages require replenishment. The combination of standard Salesforce capabilities, custom objects, controlled access, declarative automation, and Apex provides a practical implementation rather than a purely theoretical design.

5.11 Scalability & Future Plan

The current architecture can be extended by adding dashboards and reports for management, more detailed recycling and processing stages, stronger notification preferences, richer inventory analytics, and additional automation as the volume of records grows. The separation of objects and the use of focused automation components provide a foundation for these future enhancements.

END OF PROJECT DOCUMENTATION

RePlastix Innovations: Transforming Plastic Waste into Sustainable Solutions